

AI Smart Farming Robo Planner - Abstract

The AI Smart Farming Robo Planner is an intelligent agricultural automation system designed to improve farming efficiency, productivity, and resource management. The project integrates Artificial Intelligence (AI), Machine Learning, Robotics, and IoT technologies to assist farmers in performing essential agricultural tasks such as soil monitoring, irrigation planning, crop seeding, pesticide spraying, and field surveillance.

The robot continuously collects real-time data from sensors, including soil moisture, temperature, humidity, and weather conditions. Using AI-based decision-making algorithms, it analyzes the collected information and generates an optimized action plan for farm operations. The system can autonomously navigate through agricultural fields, detect obstacles, identify crop conditions, and perform tasks with minimal human intervention.

Advanced AI techniques such as search algorithms, constraint satisfaction, probabilistic reasoning, and predictive analytics are used to make intelligent decisions under uncertain environmental conditions. The robo planner helps reduce water wastage, optimize pesticide usage, minimize labor costs, and improve crop yield while promoting sustainable farming practices.

This project aims to support the concept of Smart Agriculture (Agriculture 4.0) by providing an efficient, cost-effective, and eco-friendly farming solution. The system enhances precision farming capabilities and assists farmers in making data-driven decisions for better agricultural outcomes.

Keywords: Artificial Intelligence, Smart Farming, Robotics, IoT, Precision Agriculture, Machine Learning, Automated Irrigation, Crop Monitoring, Decision Making, Sustainable Agriculture.